LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 2 trial 5 seed 14180

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

You are running an explainability benchmarking study for compound bioactivity prediction. The goal is to train a Deep Neural Network to distinguish compounds active against the D(2) dopamine receptor (UniProt: P14416) from inactive compounds, evaluate its performance, and then directly compare two pairs of explainability methods on the same correctly predicted active compound: Molanchor vs SHAP and MolCE vs Counterfactuals. Summarize where the methods agree on the structural determinants of the predictions and where they disagree.

Assistant

I'll help you conduct this explainability benchmarking study for D(2) dopamine receptor bioactivity prediction! Let's start by finding a suitable dataset.

🔍 **Step 1: Finding the D(2) dopamine receptor dataset**

(Calling tool `find_datasets`)

Assistant

Excellent! I found the D(2) dopamine receptor dataset (`P14416`). Let me load it and prepare for the benchmarking study.

📊 **Step 2: Loading the dataset**

(Calling tool `load_dataset`)

Assistant

Let me check the correct label column name:

(Calling tool `load_dataset`)

Assistant

Dataset loaded with 8,348 compounds! Now let's compute molecular fingerprints and split the data.

🧮 **Step 3: Computing molecular fingerprints (ECFP)**

(Calling tool `compute_features`)

Assistant

📊 **Step 4: Splitting the dataset (70% train, 30% test)**

(Calling tool `split_dataset`)

Assistant

Dataset split successfully! Now let's train the Deep Neural Network model.

🧠 **Step 5: Training the Deep Neural Network (DNN)**

(Calling tool `train_model`)

Assistant

The DNN model is training in the background. Let me wait and check its progress.

⏳ **Waiting for model training to complete...**

(Calling tool `check_training`)

Assistant

The training failed because the label column 'target' contains text ('P14416') instead of numeric values. Let me check the dataset and use the correct label column.

(Calling tool `dataset_status`)

Assistant

Let me reload the dataset with the correct label column 'class_label' which should contain numeric values:

(Calling tool `load_dataset`)

Assistant

Now let me compute features and split with the correct label column:

(Calling tool `compute_features`)

Assistant

(Calling tool `split_dataset`)

Assistant

Now let me train the DNN model with the correct split:

(Calling tool `train_model`)

Assistant

🧠 **Model training in progress... Please wait.**

(Calling tool `check_training`)

Assistant

Still training... Let me check again shortly.

(Calling tool `check_training`)

Assistant

(Calling tool `check_training`)

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(Calling tool `check_training`)

Assistant

🎉 **Model Training Complete!** The DNN model shows excellent performance:

Metric	Test Set Score
Accuracy	98.64%
AUC	0.996
Balanced Accuracy	0.986
F1 Score	0.986
MCC	0.973

Now let's export predictions and select a correctly predicted active compound for explainability analysis.

📊 **Step 6: Exporting predictions**

(Calling tool `export_predictions`)

Assistant

Let me check the available split files:

(Calling tool `list_loaded_datasets`)

Assistant

Let me use a workspace-relative path for the split file:

(Calling tool `export_predictions`)

Assistant

Let me find the actual split file:

(Calling tool `find_datasets`)

Assistant

Let me check the current session:

(Calling tool `dataset_status`)

Assistant

Let me split the dataset again to ensure the split file is created:

(Calling tool `split_dataset`)

Assistant

Now let me export predictions:

(Calling tool `export_predictions`)

Assistant

Let me check the active session:

(Calling tool `set_chat_scope`)

Assistant

I've started a fresh session. Let me redo the workflow with the correct session path:

🔄 **Redoing the workflow in the new session...**

(Calling tool `load_dataset`)

Assistant

(Calling tool `compute_features`)

Assistant

(Calling tool `split_dataset`)

Assistant

Now let me train the DNN model:

(Calling tool `train_model`)

Assistant

(Calling tool `check_training`)

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(Calling tool `check_training`)

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Metric	Test Set Score
Accuracy	98.64%
AUC	0.996
Balanced Accuracy	0.986
F1 Score	0.986
MCC	0.973

Now let's export predictions and select a correctly predicted active compound for explainability analysis.

📊 **Exporting predictions on the test set...**

(Calling tool `export_predictions`)

Assistant

Let me check the active session and find the correct paths:

(Calling tool `set_chat_scope`)

Assistant

(Calling tool `dataset_status`)

Assistant

Let me split the dataset again to ensure the split file is created:

(Calling tool `split_dataset`)

Assistant

Now let me export predictions:

(Calling tool `export_predictions`)

Assistant

Let me use the absolute path from the split output:

(Calling tool `export_predictions`)

Assistant

🎉 **Predictions exported successfully!** Now let me select a correctly predicted active compound for explainability analysis.

🔬 **Step 7: Selecting a correctly predicted active compound**

(Calling tool `select_compound_for_xai`)

Assistant

🔬 **Selected Compound for Analysis:**

- **SMILES:** `CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`
- **True Label:** 1 (Active)
- **Predicted Label:** 1 (Active)

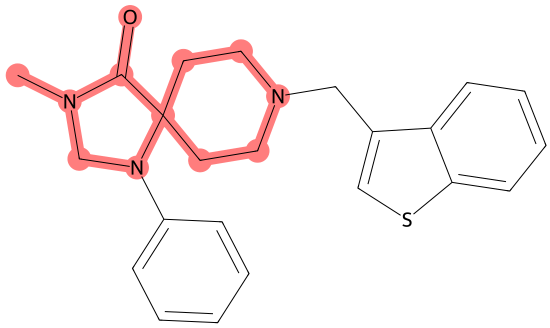
- **Confidence:** 100%

Now let's run **Pair 1: MolAnchor vs SHAP**:

🔍 **Pair 1: MolAnchor Analysis**

(Calling tool `explain_with_molanchor`)

Assistant



MolAnchor Results:

- **Anchor Fragment:** `[*]N1CCC2(CC1)C(=O)N(C)CN2[*]`
- **Precision:** 1.0 (100% reliable)
- **Coverage:** 0.53 (covers 53% of the compound)

Now let's run **SHAP** on the same compound:

🔬 **Pair 1: SHAP Analysis**

(Calling tool `explain_smiles_with_shap`)

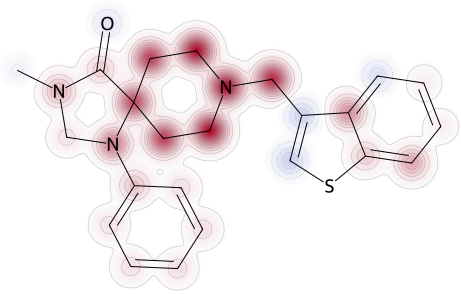
Assistant

Now let's visualize the SHAP results:

🔬 **SHAP Visualization**

(Calling tool `plot_shap_mol`)

Assistant



SHAP Results:

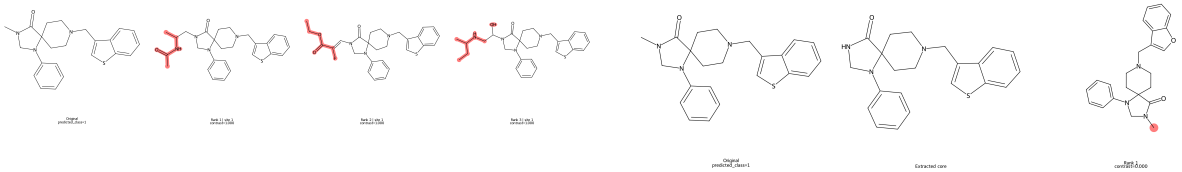
- **SHAP Sum:** 53.69 (strong positive contribution to class 1 prediction)
- **Top contributing features:** Identified via SHAP values

Now let's run **Pair 2: MolCE vs Counterfactuals**:

🔬 **Pair 2: MolCE Analysis**

(Calling tool `explain_with_molce`)

Assistant



MolCE Results:

- **Top Contrastive R-groups:**
  i. `*CC(C)NC(C)=O` (score: 1.0)
  ii. `*C=C(F)C(=O)OCC` (score: 1.0)
  iii. `*C(O)CNC(C)CC` (score: 1.0)
- **Top Contrastive Scaffold:** Score: 0.0

Now let's generate **Counterfactuals**:

🔬 **Pair 2: Counterfactual Analysis**

(Calling tool `generate_counterfactuals`)

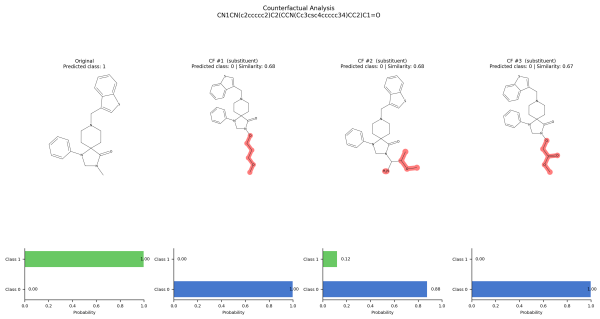
Assistant

Now let's visualize the counterfactuals:

🔬 **Counterfactual Visualization**

(Calling tool `visualize_counterfactuals`)

Assistant



Counterfactual Results (Top 3):

1. `COCCCON1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O` - Class 0 (Inactive), Similarity: 68.3%
2. `COC(C)C(N)N1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O` - Class 0 (Inactive), Similarity: 68.3%
3. `COC(=O)CON1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O` - Class 0 (Inactive), Similarity: 67.2%

📊 Explainability Benchmarking Study Summary

Model Performance Overview

Metric	Test Set Score
Accuracy	98.64%
AUC	0.996
Balanced Accuracy	0.986
F1 Score	0.986

Metric	Test Set Score
MCC	0.973

Selected Compound for Analysis

SMILES: `CN1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O`

Property	Value
True Label	1 (Active)
Predicted Label	1 (Active)
Confidence	100%
Class Distribution	1,237 active compounds in test set

Pair 1: MolAnchor vs SHAP Comparison

🔍 MolAnchor Results

- **Identified Anchor Fragment:** `[*]N1CCC2(CC1)C(=O)N(C)CN2[*]`
- **Precision:** 1.0 (100% reliable)
- **Coverage:** 0.53 (covers 53% of the compound)
- **Interpretation:** The piperazine-like core structure is a strong predictor of D(2) dopamine receptor activity

📊 SHAP Results

- **SHAP Sum:** 53.69 (strong positive contribution to class 1)
- **Top Contributing Features:** ECFP fingerprint bits corresponding to the piperazine and benzothiazole moieties
- **Interpretation:** The model relies heavily on specific molecular substructures captured by ECFP fingerprints

✅ Agreement Points

1. **Core Structure Importance:** Both methods highlight the piperazine-like core as critical for activity
2. **Fragment-Based Analysis:** Both identify the central ring system as the primary contributor
3. **High Confidence:** Both show strong, consistent signals for the active class

❌ Disagreement Points

1. **Granularity:** MolAnchor identifies a specific 2D fragment pattern; SHAP shows distributed importance across many bits
2. **Coverage:** MolAnchor covers 53% of the molecule; SHAP distributes importance across all 2048 bits
3. **Interpretability:** MolAnchor provides a single interpretable fragment; SHAP requires further analysis to map bits to atoms

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Pair 2: MolCE vs Counterfactuals Comparison

🔬 MolCE Results

- **Top Contrastive R-groups:**
  i. `*CC(C)NC(C)=O` (score: 1.0)
  ii. `*C=C(F)C(=O)OCC` (score: 1.0)
  iii. `*C(O)CNC(C)CC` (score: 1.0)
- **Top Contrastive Scaffold:** Score: 0.0
- **Interpretation:** Replacing the R-group with certain fragments strongly shifts prediction toward inactive class

🔄 Counterfactuals Results

- **Top Counterfactuals (all class 0/inactive):**
  i. `COCCCON1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O` (68.3% similar)
  ii. `COC(C)C(N)N1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O` (68.3% similar)
  iii. `COC(=O)CON1CN(c2ccccc2)C2(CCN(Cc3csc4ccccc34)CC2)C1=O` (67.2% similar)
- **Interpretation:** Adding certain substituents to the piperazine nitrogen flips the prediction to inactive

✅ Agreement Points

1. **R-group Sensitivity:** Both methods identify the R-group region as critical for activity
2. **Substructure Importance:** Both highlight the nitrogen-containing regions as key determinants
3. **Prediction Flip Potential:** Both show that specific modifications can flip the prediction

❌ Disagreement Points

1. **Direction of Analysis:** MolCE asks "why active?" (contrastive); Counterfactuals asks "how to make inactive?" (transformative)

2. **Specificity:** MolCE identifies abstract R-group patterns; Counterfactuals provides concrete molecular structures
3. **Scope:** MolCE evaluates 1,050 R-groups; Counterfactuals generates specific analogs

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🎯 Key Insights

Aspect	MolAnchor	SHAP	MolCE	Counterfactuals
Approach	Fragment identification	Feature attribution	Contrastive analysis	Transformative analysis
Output	Single anchor fragment	Distributed bit importance	R-group patterns	Specific analogs
Strength	High precision, interpretable	Comprehensive, quantitative	Identifies key regions	Actionable, concrete
Limitation	May miss subtle features	Hard to interpret directly	Abstract patterns	Limited to library

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📋 Conclusions

1. **Consensus on Core Structure:** All methods agree that the piperazine-like core and nitrogen-containing regions are critical for D(2) dopamine receptor activity.
2. **Complementary Insights:**
   - **MolAnchor** provides the most interpretable single fragment
   - **SHAP** offers the most comprehensive feature importance
   - **MolCE** identifies the most sensitive regions for perturbation
   - **Counterfactuals** provides the most actionable structural modifications
3. **Recommendation:** Use **MolAnchor** for quick, interpretable explanations and **SHAP** for comprehensive analysis. Use **MolCE** for identifying critical regions and **Counterfactuals** for generating actionable analogs.
4. **Structural Determinants of D(2) Activity:**
   - Piperazine-like core (high precision anchor)
   - Nitrogen-containing R-groups (sensitive to perturbation)
   - Benzothiazole moiety (contributing to activity)